

# IX | ELEC70126 - Internet of Things and Applications

## Module Exam

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|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Module code and Name</b>                       | Nocturnal Microclimate and Sleep Architecture: A personal IOT study                                                                                                     |
| <b>Student CID</b>                                | 06048874                                                                                                                                                                |
| <b>Assessment date</b>                            | 2pm 25/03/2026                                                                                                                                                          |
| <b>Code &amp; Data (publicly accessible link)</b> | <a href="https://github.com/ic-yidanWang/IOT_final_project">https://github.com/ic-yidanWang/IOT_final_project</a>                                                       |
| <b>Platform link (publicly accessible link)</b>   | <a href="https://ic-yidanwang-iot-final-project-step6-dashboard-1va1uq.streamlit.app/">https://ic-yidanwang-iot-final-project-step6-dashboard-1va1uq.streamlit.app/</a> |

# Nocturnal Microclimate and Sleep Architecture: A personal IOT study

## 1 Sensing

### 1.1 Introduction and Objectives

#### 1.1.1 Introduction

Sleep is one of the most consequential biological processes governing human health, yet it remains almost entirely unmonitored in everyday life. Most people have no systematic record of their sleep quality, let alone any understanding of why it varies night to night. Personal IoT sensing is positioned to close this gap. The relationship between thermal environment and sleep is well established. Sleep initiation is closely coupled to a circadian decline in core body temperature, with the surrounding thermal environment serving as the primary medium for heat dissipation Harding et al. (2019). Elevated ambient temperature or humidity impairs this thermoregulatory process, delaying sleep onset and reducing slow-wave sleep. Yet this research has almost exclusively been conducted in climate-controlled polysomnography labs bearing little resemblance to a real bedroom. What happens when the bedroom is not controlled, when outdoor weather changes nightly, heating switches off at midnight, or a wet week raises indoor humidity? Almost no longitudinal personal data exists on how this natural microclimate variability relates to real sleep architecture. Until recently, measuring both the sleep environment and biological response simultaneously required either a hospital lab or prohibitively expensive equipment. The convergence of low-cost embedded sensing, cloud meteorological APIs, and consumer wearables has changed this. The resulting data is noisy, heterogeneous, and statistically sparse, but it opens a previously inaccessible question: does real-world nocturnal microclimate leave a measurable imprint on sleep architecture? This project attempts to answer that question, or at minimum, to characterise exactly how much evidence a personal IoT study of this scale can and cannot provide.

#### 1.1.2 Project Objectives

This study aims to explore a core question: Do natural and uncontrolled changes in the nighttime microclimate of a domestic bedroom over a 14-night span have measurable effects on sleep structure? To answer this question, it needs to be broken down into four analytical objectives. The first is to quantify bivariate correlations between nightly environmental variables (indoor temperature, indoor humidity, and outdoor meteorolog-

ical conditions) and sleep quality metrics, including sleep score, deep sleep percentage, and REM proportion. Before that, a normality test is needed to determine whether the parametric or nonparametric method is more appropriate. The second concerns temporal structure. Thermoregulatory responses operate on timescales of minutes, meaning a change in room temperature may only manifest in a sleep stage change some time later. A within-night lag cross-correlation analysis therefore tests at which lag  $k$  any association peaks. This speaks directly to the plausibility of a causal mechanism. The third asks not just whether the microclimate affects sleep, but which aspect of sleep it affects. Deep slow-wave sleep, REM, and light NREM are physiologically distinct and differentially sensitive to environmental perturbation. The study therefore compares environmental correlations across sleep stage proportions, including deep sleep, REM, and core sleep, to examine whether any association is localised to a specific stage or distributed across the whole night. Finally, the study asks how much its findings can be trusted. A 14-night single-subject dataset imposes hard limits on statistical inference. Power analysis is therefore applied to characterise, for each effect size of interest, the probability that a true effect would have been detected.

## 1.2 Data Sources and sensing set-up

The study draws on three data sources capturing the two sides of the research question, the physical environment and the biological response: an indoor microclimate sensor, an outdoor meteorological API, and a consumer wearable sleep staging device.

### 1.2.1 Indoor Microclimate — Arduino Nano and DHT11

Indoor temperature and relative humidity were measured using a DHT11 sensor connected to an Arduino Nano, mounted at approximately 1.0 m height and 1.5 m from the nearest window. No heating or air conditioning was operated during any of the 14 study nights, making indoor temperature a function of outdoor conditions, building thermal mass, and occupant metabolic output alone. The DHT11 measures temperature (0–50 °C,  $\pm 2$  °C, 1 °C resolution) and relative humidity (20–90 % RH,  $\pm 5$  % RH, 1 % RH resolution) at a 15-minute sampling interval. The  $\pm 2$  °C accuracy is non-trivial relative to the thermoregulatory gradients relevant to sleep onset (0.5–1 °C), so the sensor resolves coarse nightly trends but not fine transient fluctuations. This motivated the use of nightly mean temperature as the primary predictor, as averaging 30 readings per night substantially reduces effective measurement noise.

### 1.2.2 Outdoor Meteorology — OpenWeatherMap API

Outdoor conditions were sourced from the OpenWeatherMap Current Weather API at 15-minute intervals for the study location in London (51.5074 °N, 0.1278 °W). Four vari-

ables were extracted: outdoor temperature ( $^{\circ}\text{C}$ ), relative humidity (% RH), precipitation (mm/h), and wind speed (m/s).

### 1.2.3 Sleep Staging — Apple Watch Series 9

Sleep staging data were collected via Apple Watch Series 9 worn throughout each night, which classifies sleep into Deep, Core, REM, and Awake states. As with all consumer wearables, stage classifications carry some measurement uncertainty relative to clinical polysomnography, particularly for deep sleep detection. However, because any systematic sensor bias remains approximately constant across nights, night-to-night relative differences in stage proportions are more reliable than absolute durations.

## 1.3 Data collection and storage/communications process

The three pipelines follow distinct communication pathways before converging at a daily manual integration step. As shown in Figure 1.

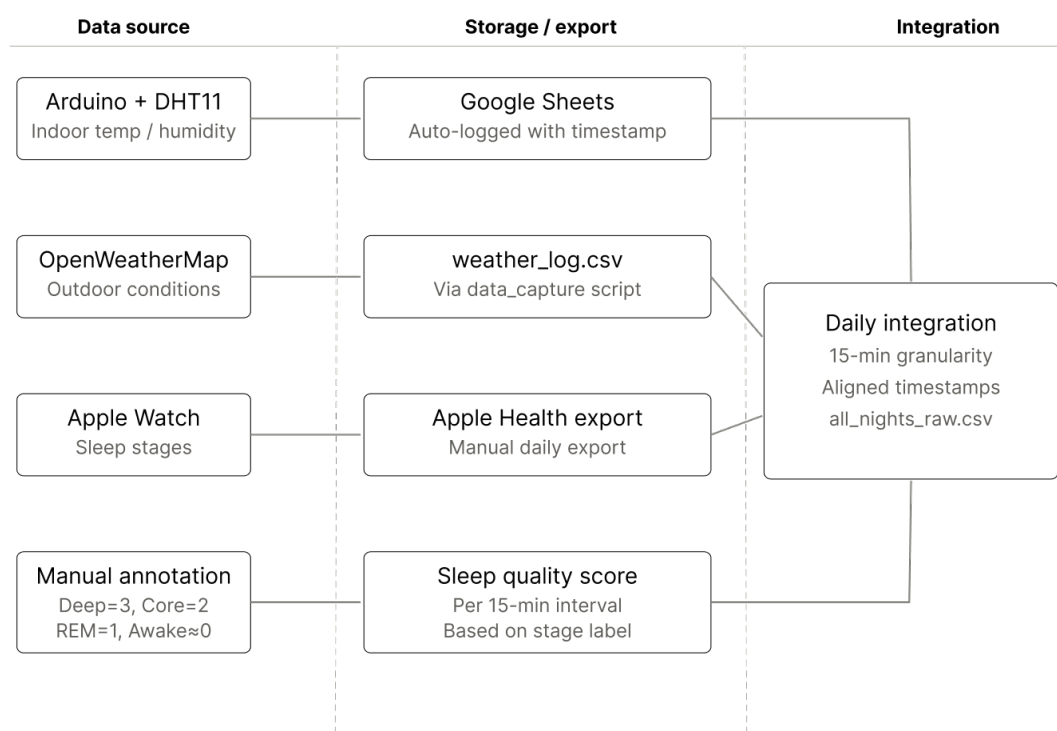


Figure 1: End-to-end data flow from three sensing pipelines to unified dataset.

### 1.3.1 Indoor Pipeline — Arduino to Google Sheets

Arduino Nano reads the DHT11 temperature sensor every 15 minutes and transmits each reading in the format of T=22.5, H=47.0 through the serial port (115,200 baud). The host

script forwards these values to Google Sheets, and Google Sheets will add a server-side timestamp to generate table rows in the format of "timestamp | temperature | humidity". In order to ensure that the data can be preserved permanently after the interruption of the host session at night, we choose to use cloud storage instead of local logging. The main limitation is that the disconnection of WiFi will cause the silent failure of the whole process; and the production system will cache the records locally and synchronise them after reconnecting. At the end of each night, we will manually export the record and merge it into the CSV file of the night.

### **1.3.2 Outdoor Pipeline — API Polling to Local Storage**

Outdoor meteorological data were collected by polling the OpenWeatherMap Current Weather API at 15-minute intervals throughout each night. Four fields were extracted per call: outdoor temperature, relative humidity, precipitation, and wind speed, each timestamped using a unified system clock to ensure temporal consistency during daily integration. If the API fails, the script will jump to the next scheduled poll, resulting in missing data instead of reusing outdated values. During the preprocessing process, the relevant data window of each study night is extracted through the time stamp.

### **1.3.3 Sleep Pipeline — Apple Health Export and Manual Staging**

At the end of each night, sleep stage data were exported from Apple Health. Apple Watch records stage transitions at 30-second resolution internally but exports them as labelled intervals. These were manually converted to a 15-minute epoch grid by assigning each window the stage occupying its majority. The sleep score used here is derived directly from Apple Watch's stage classification (REM=1, Core=2, Deep=3), making it a weighted mean of stage composition. Awake periods were excluded as all such events lasted one to two minutes, which was shorter than a single epoch and could not be meaningfully represented at 15-minute resolution. This is an alignment decision. Nightly sleep score and total duration from Apple Health were appended as constant values across all epochs of that night.

### **1.3.4 Daily Integration and Preprocessing**

The three data streams every night are manually aligned to the shared 15-minute timestamp grid and merged into a separate CSV file that night. Due to the slight inter-source clock deviation in the Google Sheets server timestamp, system clock and Apple Health's internal clock, we solve this deviation by aligning all records to the nearest 15-minute boundary, and the estimated maximum deviation is  $\pm 2$  Minutes. Then, we connected all 14 files that night and added derivative columns for recording the date, time (minutes) and numerical stage coding to generate a summary table for subsequent analysis.

## 1.4 Basic Characteristics of the End-to-End Data Acquisition Setup

The complete data set covers 14 consecutive nights (February 26 to March 11, 2026), generating 385 cycle-level records. The number of cycles per night is between 20 and 36 (average: 27.5), reflecting the natural fluctuation of sleep time, with an average sleep time of 6.875 hours per night. One night (3 March) experienced a three-hour gap in sensor data due to a pipeline interruption, resulting in 12 missing environmental epochs. The affected variables (indoor temperature, indoor humidity, outdoor temperature, outdoor humidity, wind speed, and precipitation) were recovered via degree-2 polynomial interpolation fitted to the known records of that night, thus retaining the time trend in the session without exceeding the observation range.

## 2 Internet of Things

### 2.1 Data Interaction/visualisation/actuation platform

The interactive dashboard was developed using Streamlit and serves as the central interface for data exploration across the study period (26 February – 11 March 2026,  $n = 14$  nights). It is organised into five pages: Overview presents a nightly summary table alongside cross-night trends in sleep duration, sleep score, stage composition, and both indoor and outdoor temperature and humidity; Night Explorer allows per-night inspection of the full sleep hypnogram overlaid with real-time indoor temperature, indoor humidity, outdoor temperature, and wind speed traces; Correlation provides an interactive Spearman correlation matrix and scatter plot for any environmental–sleep variable pair, as well as a lag cross-correlation chart showing the temporal relationship between environmental signals and sleep stage depth; Sleep Structure displays the pooled stage-transition probability matrix, per-night sleep efficiency metrics (Deep/Core/REM minutes, cycle count, and latency to first REM and Deep), and a stacked stage-composition chart across all nights; and Power Analysis presents sample-size sensitivity curves to contextualise the statistical limits of the dataset. Together, these pages allow a complete walkthrough of the data from raw observation to inferential context without requiring any manual re-execution of analysis scripts.

As a further design consideration, if the dashboard is connected to the smart home actuator through MQTT, the real-time humidity threshold detected in the associated page can directly trigger the dehumidifier, thus forming a complete perception-inference-action cycle.

Before correlation analysis, we built a descriptive visualisation process to characterise the data set. The nocturnal change curve of all five environmental variables shows that

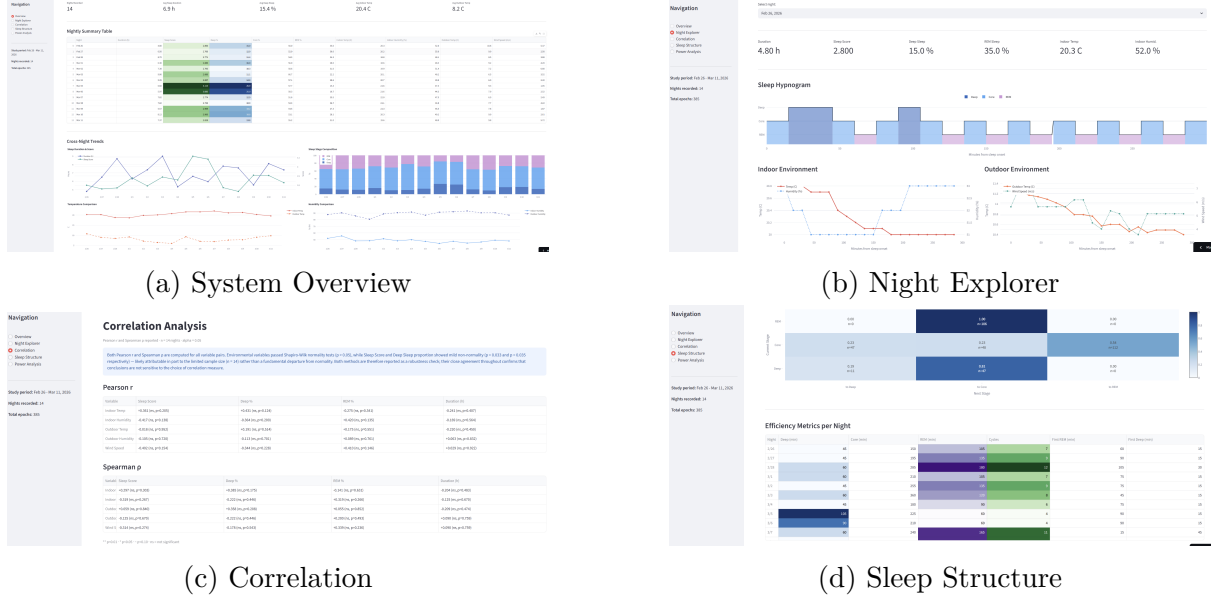


Figure 2: Visualisation of selected dashboard features

the indoor temperature remains relatively stable for 14 nights (18.6–22.0°C), while the night fluctuations of outdoor temperature and wind speed are significantly greater. Box-plot distributions confirmed that indoor humidity (44–56% RH) and sleep score occupied narrow ranges. Normality was assessed via Shapiro-Wilk tests: although all environmental variables passed ( $p > 0.05$ ), Sleep Score and Deep Sleep proportion showed mild non-normality ( $p = 0.033$  and  $p = 0.035$  respectively). Given these violations and the small sample size ( $n = 14$ ), Spearman  $\rho$  is adopted as the primary correlation measure throughout. Pearson  $r$  is reported alongside as a robustness check; the close agreement between the two measures confirms that conclusions are not sensitive to the choice of method. Sleep-stage hypnograms were plotted for each night with simultaneous indoor temperature and humidity overlaid, providing a qualitative sense of how environmental conditions co-vary with stage transitions within individual nights, and cross-night summary trends of key sleep and environmental variables were compiled as a navigational reference for the analyses that follow.

## 2.2 Data Analytics, Inferences and Insights

Cross-night Spearman and Pearson correlations were computed between all environmental predictors, sleep score, and sleep duration. As shown in the correlation matrix (Fig. 3), no environmental variable reached statistical significance against either sleep outcome (all  $p > 0.20$ ). The strongest directional trends against sleep score were indoor humidity (Pearson  $r = -0.42$  / Spearman  $\rho = -0.32$ ) and wind speed ( $r = -0.40$  /  $\rho = -0.31$ ), both negative, while indoor temperature showed a modest positive trend ( $r = +0.36$  /  $\rho = +0.30$ ). Among the environmental variables themselves, outdoor temperature and

outdoor humidity exhibited a significant inter-correlation ( $\rho = -0.72$ ,  $p < 0.01$ ), physically expected as colder nights tend to carry higher relative humidity. Sleep duration showed no meaningful association with sleep score ( $\rho = -0.25$ ).

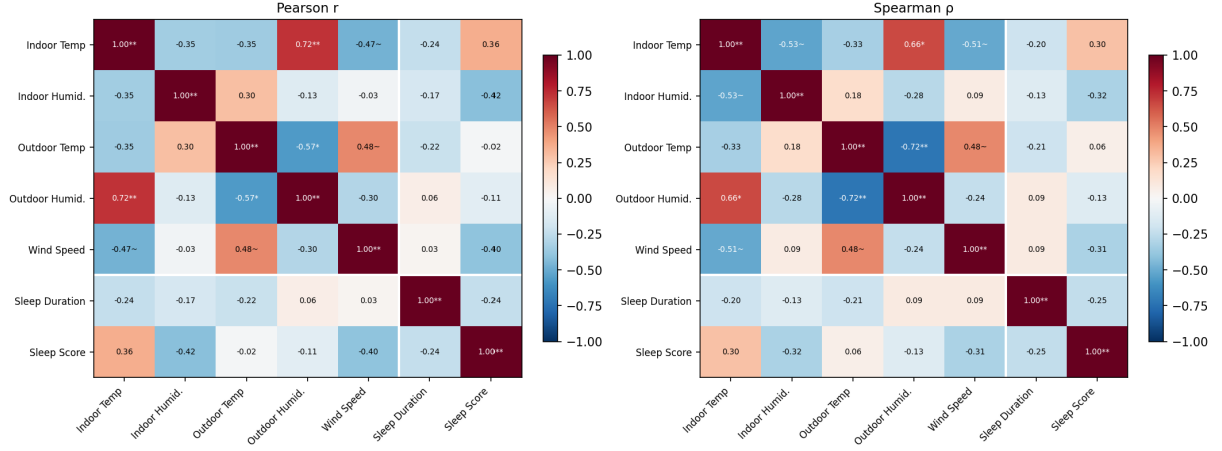


Figure 3: Nightly Correlation Matrix(n = 14 nights)

Breaking the sleep outcome down by stage, linear scatter analyses against sleep score, deep sleep proportion, and REM proportion reveal a consistent pattern: REM shows near-zero correlations with all environmental variables (all  $|\rho| \leq 0.34$ ), whereas sleep score and deep sleep proportion show moderate though non-significant trends with indoor humidity ( $\rho = -0.32$ ) and indoor temperature ( $\rho = +0.38$  against deep sleep). This suggests that whatever environmental influence may exist operates at the level of overall sleep quality and slow-wave sleep rather than through REM architecture.

The most informative signal emerged at the within-night timescale. Lag cross-correlation analysis revealed that outdoor temperature was the only variable to show a peak in the positive lag region (+15 minutes, mean Pearson  $r = +0.31$ , Fig. 4), consistent with the direction one would expect if environmental change precedes sleep stage response, plausibly reflecting the time required for outdoor thermal conditions to propagate indoors and engage thermoregulatory mechanisms. This pattern is physiologically interpretable, but given the small sample size, it should be treated as a suggestive signal rather than a confirmatory finding. Indoor temperature, by contrast, peaked at -45 minutes ( $r = +0.22$ , Fig. 4), suggesting the reverse temporal ordering. This is more plausibly attributed to a structural confound: deep sleep concentrates in the first half of the night, which also tends to be the warmer half as indoor temperature drifts downward across the night, producing a spurious negative lag regardless of any true causal relationship. With only 14 nights of data, this interpretation remains speculative. A per-night Spearman analysis further showed that indoor temperature was positively associated with sleep stage depth on the majority of individual nights, a signal that cross-night aggregation into nightly means suppresses entirely.

At the cross-night level, an exploratory OLS regression with nightly mean sleep score



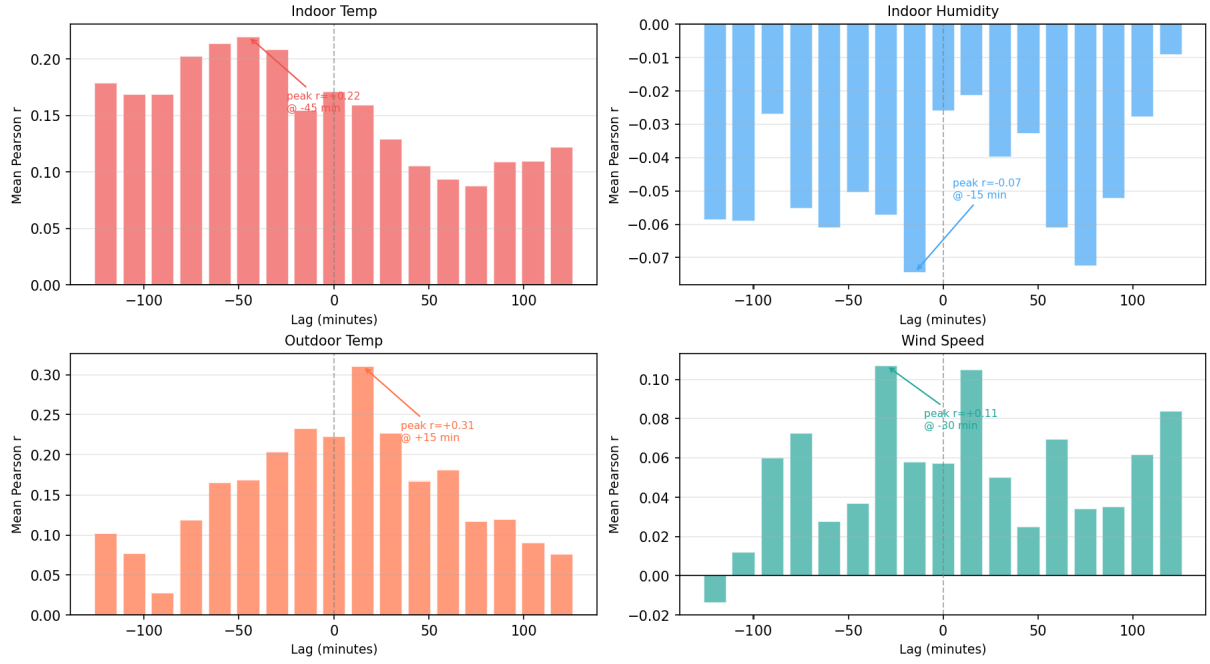


Figure 4: Lag Cross-Correlation: Environment Variables -> Sleep Stage

as the outcome was conducted. The full model achieved  $R^2=0.50$  but adjusted  $R^2=0.27$ , with the gap reflecting overfitting, and multicollinearity among predictors further destabilised individual coefficients. Ridge regression with LOOCV was therefore applied as a more conservative alternative, with the shrinkage path identifying outdoor temperature as the most stable positive predictor, with indoor humidity and wind speed as the most stable negative predictors. All regression results are treated as hypothesis-generating only.

As a further exploratory step, indoor-outdoor differential variables were constructed to test whether it has effect on sleep quality. This yielded no meaningful associations with any sleep outcome (all Spearman  $\rho \leq 0.18$ , all  $p > 0.50$ ), and this line of analysis was not pursued further.

## 2.3 Discussions on the important aspects of the project

The central statistical limitation of this study is its sample size. As shown in the power analysis, at  $n = 14$  nights the study is adequately powered (power  $\geq 0.80$ ) only to detect large correlations ( $|\rho| \geq 0.70$ ). Indoor humidity against sleep score at  $\rho = -0.32$ , indoor temperature against deep sleep at  $\rho = +0.38$  both in the medium range, for which achieved power is approximately 0.45. This means that even if these relationships are real, the current dataset has only a 45% chance of detecting them as statistically significant. The absence of  $p < 0.05$  results in Section 2 should therefore not be interpreted as evidence of no effect; rather, the study is best characterised as hypothesis-generating, with the directional trends warranting confirmation in a larger sample.

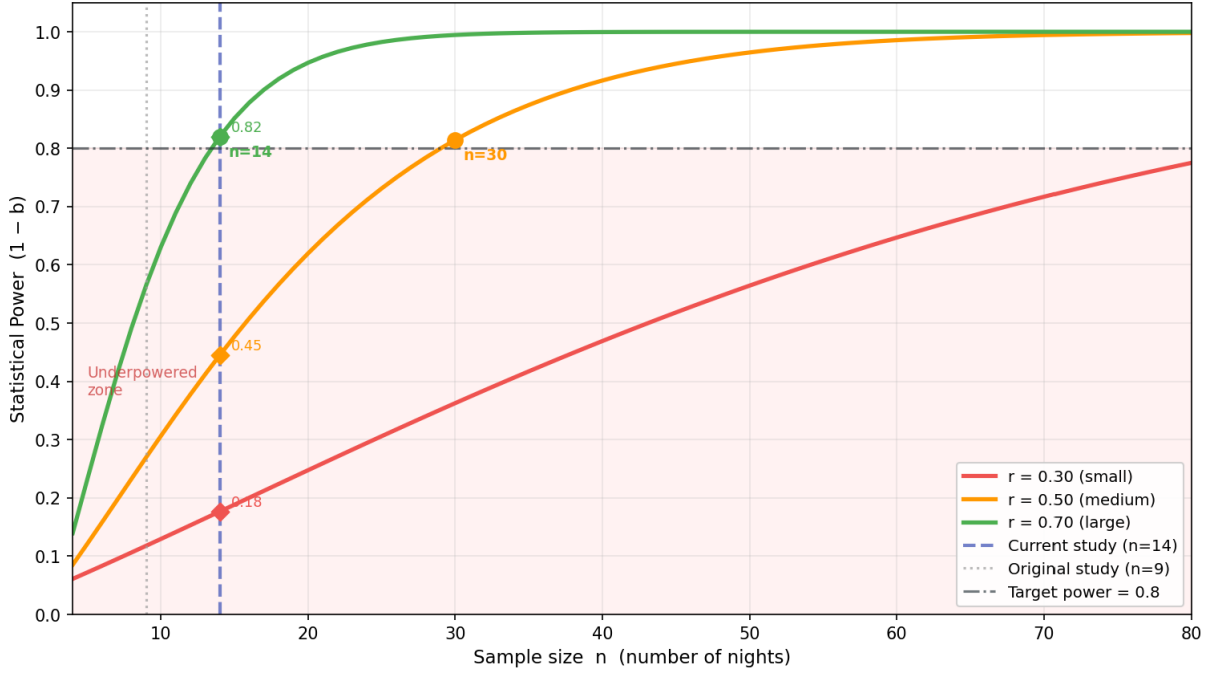


Figure 5: Power-Sample Size Sensitivity Curve (Spearman  $\rho$ ,  $\alpha = 0.05$ )

Beyond sample size, several uncontrolled confounders may account for a portion of the cross-night variance in sleep quality. Daily physical activity level, which is known to promote slow-wave sleep, was not recorded and could co-vary with environmental conditions on any given night. Bedtime consistency was similarly uncontrolled; as a single-subject dataset collected during an active academic term, sleep onset times varied considerably with coursework deadlines, study load, and daily schedule. Factors may independently influence both sleep architecture and the duration of environmental exposure within any given night. Stress and cognitive arousal associated with academic pressure represent a particularly plausible confound, given their known suppressive effect on slow-wave sleep, and cannot be separated from the environmental signal in the current design.

Data quality is also a constraint. The environmental sensors were also fixed at a single room location. Bed-level temperature and humidity were not directly measured, introducing a spatial gap between the recorded exposure and the actual situa.

## 2.4 Avenues for future work and potential impact

The first task of future work is to expand the data set. As described in Section 2.3, it takes about 30 nights to reliably detect medium-scale environmental impacts. Extending the study to this threshold and recording previously uncontrolled variables, such as daily steps, bedtime and air conditioning usage, will help to conduct partial correlation analysis, thus distinguishing microclimate signals from lifestyle confounding factors. In addition, intra-control intervention in subjects, that is, deliberately maintaining indoor humidity at different target levels at alternating nights, will help to make causal inferences that

cannot be supported by observational data alone.

The sensor infrastructure also needs to be improved. Replacing the fixed room sensor with a sensor installed by the bed can narrow the spatial gap between the recorded microclimate and the actual microclimate. Deploying multiple sensors in different positions in the room can draw a spatial distribution map of the temperature gradient, which may be more physiologically relevant than the single-point average.

Transcending a single research object is the most important way to achieve the universality of results. A multiparticipant study covering different housing types, climate and demographic characteristics will help reveal whether the observed relationship between humidity and sleep is specific or universally applicable. Any such expansion will cause important privacy and security issues. Sleep staging and biometric data are one of the most sensitive categories of personal health information: as shown in the Internet of Things security literature, even anonymous behavioural data flows can be re-identified through traffic pattern analysis or combined with auxiliary data sources. Therefore, data collection should follow the principle of minimisation, collecting only the data necessary for analysis and the original sensor data flow should be processed locally as much as possible instead of being transmitted to the cloud, so as to reduce the attack surface of privacy leakage and interception. Informed consent, transparent data retention policies and clear management of third-party access are prerequisites for any deployment beyond the scope of a single self-research.

Finally, the dashboard architecture developed by this project provides a natural foundation for the closed-loop system. In this system, when the humidity reading exceeds the preset threshold, the intelligent dehumidifier can be triggered through the MQTT protocol, and the sleep score of the night can be used as feedback to evaluate the intervention effect. Any large-scale deployment, especially in the clinical environment, requires regulatory approval and ongoing audit. In a highly health-sensitive environment, automated operations will introduce additional failure patterns, including denial-of-service attacks and threshold calibration errors, which must be addressed through security design principles rather than treated as afterthoughts.

## References

- Harding, E. C., Franks, N. P. and Wisden, W. (2019), ‘The temperature dependence of sleep’, *Frontiers in Neuroscience* **13**, 336. PMCID: PMC6491889.  
**URL:** <https://doi.org/10.3389/fnins.2019.00336>